SWEP SOFTWARE ENGINEERING PROJECT • PRODUCTION RELEASE

CampusGo

Smart Campus Transportation Management System for Obafemi Awolowo University (OAU)

A high-performance digital coordination platform designed to eliminate commuter uncertainty, provide verified real-time bus availability across campus parks, manage tricycle charters with transparent fare estimation, and deliver multi-tiered administrative governance with automated disciplinary workflows.

LIVE DEPLOYED URL https://campus-go-7fcb.onrender.com	GITHUB REPOSITORY Bukola-12/Smart-transportation-system
TARGET CAMPUS Obafemi Awolowo University, Ile-Ife (OAU)	KEY OPERATING NODES Campus Gate Car Park & SUB Car Park
USER ROLES Student, Driver, Park Admin, Central Admin	SPECIFICATION VERSION Version 1.0 (Live Deployed)

1. Executive Summary & System Scope

Campus transportation across large university campuses such as Obafemi Awolowo University faces systemic challenges: students often do not know whether a bus is loading at Campus Gate or SUB Car Park, when the last vehicle departed, or whether availability records are fresh. Concurrently, tricycle charters have traditionally been coordinated via informal, unverified roadside negotiations.

CampusGo bridges this operational gap by introducing a lightweight, real-time coordination layer that complements the university's physical ticket system without requiring expensive GPS hardware.

1.1 Core Boundaries & Guarantees

Complementary Ticketing Model (FR-23)

GUARANTEED BOUNDARY

CampusGo explicitly serves as an availability and coordination tool. It does *not* claim to replace physical ticketing booths or collect bus ticket fares. Students are clearly informed that bus tickets continue to be acquired through established park ticket stands.

Honest Availability & Zero GPS Pretense (FR-24)

ACCURACY PRINCIPLE

Rather than rendering unreliable, simulated GPS map pins, CampusGo reports honest operational statuses backed by verifiable timestamps and freshness indicators (e.g., "Confirmed 4 mins ago"). Records older than 2 hours are automatically flagged as **Unconfirmed** (FR-09).

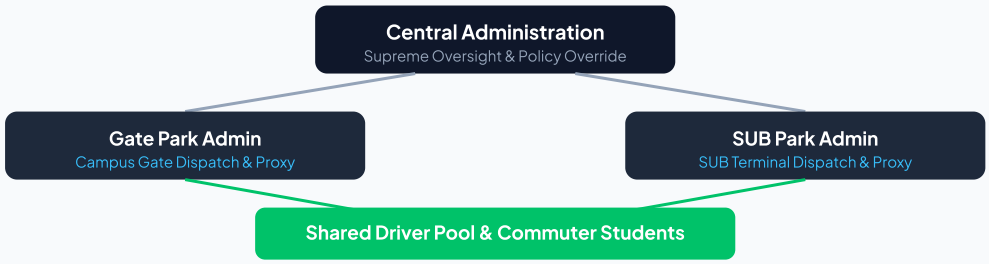
Unified Shared-Fleet Architecture (FR-14)

FLEET CONTINUITY

Vehicles and drivers belong to a single university-wide fleet pool that moves dynamically between Campus Gate Car Park and SUB Car Park. Moving between parks does not create duplicate driver records or split inventory.

1.2 System Architecture & Role Topology

FIGURE 1: CAMPUSGO 4-TIER ROLE & FLEET DISTRIBUTION MODEL

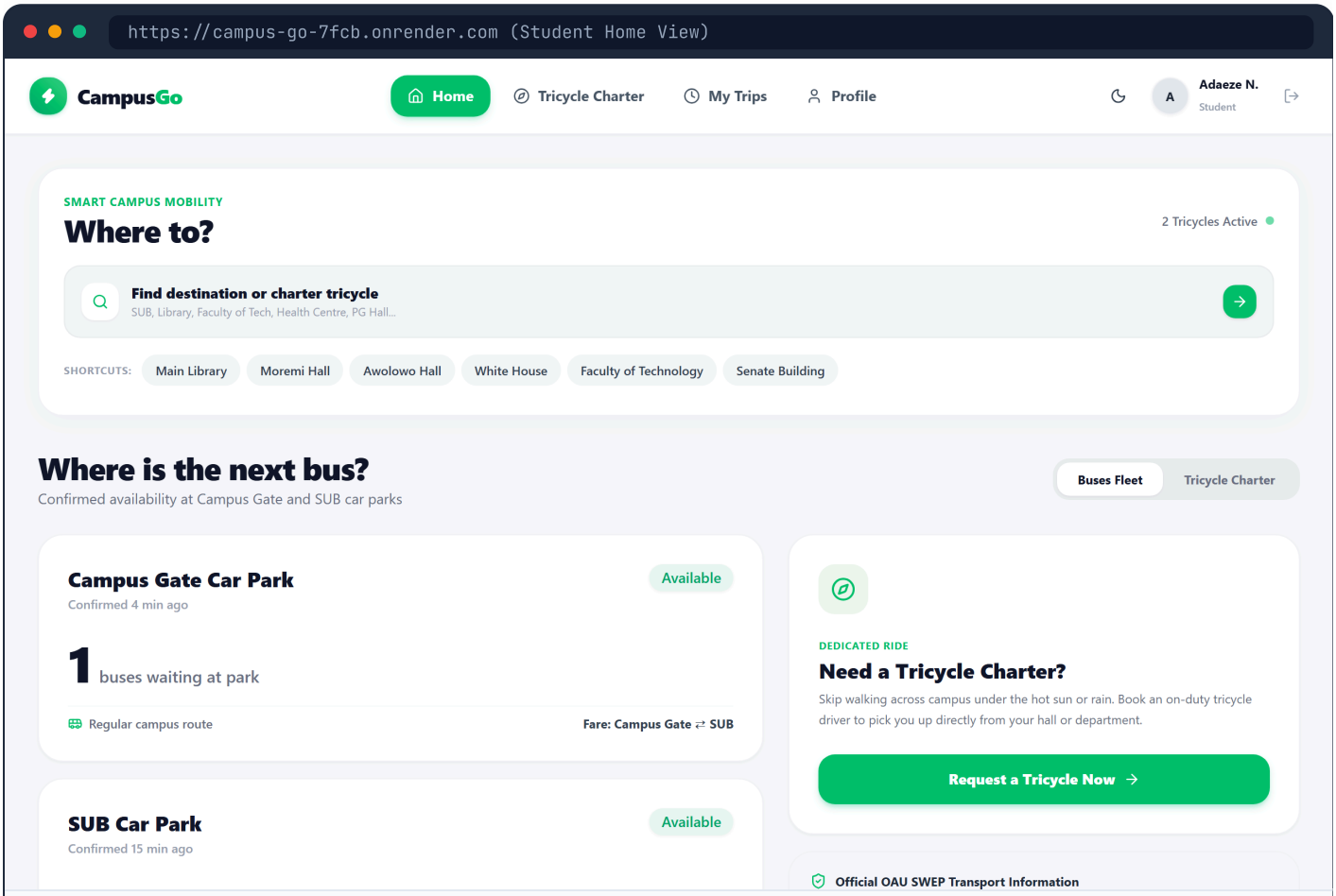


2. Student Portal Core Functionalities

The student portal is crafted with a mobile-first, Bolt-inspired design language allowing students to check live availability in seconds and submit on-demand tricycle charter requests with upfront price estimates.

2.1 Verified Bus Availability Monitor (FR-20, FR-21, FR-22)

Dual-Park Live Overview: Side-by-side availability counters for Campus Gate Car Park and Student Union Building (SUB) Car Park.
Color-Coded Freshness Badges: Green (*Confirmed < 30 mins ago*), Amber (*Aging > 1 hour*), and Muted (*Unconfirmed > 2 hours*).
Fleet State Breakdown: Live counts of buses currently loading, buses en route between parks, and standby vehicles.



Screenshot 1: Student Live Bus Availability Interface featuring live park counters, freshness badges, and quick tricycle charter button.

2.2 Bolt-Inspired Tricycle Chartering Workflow (FR-25 to FR-30)

Intelligent Campus Location Selector (FR-25)

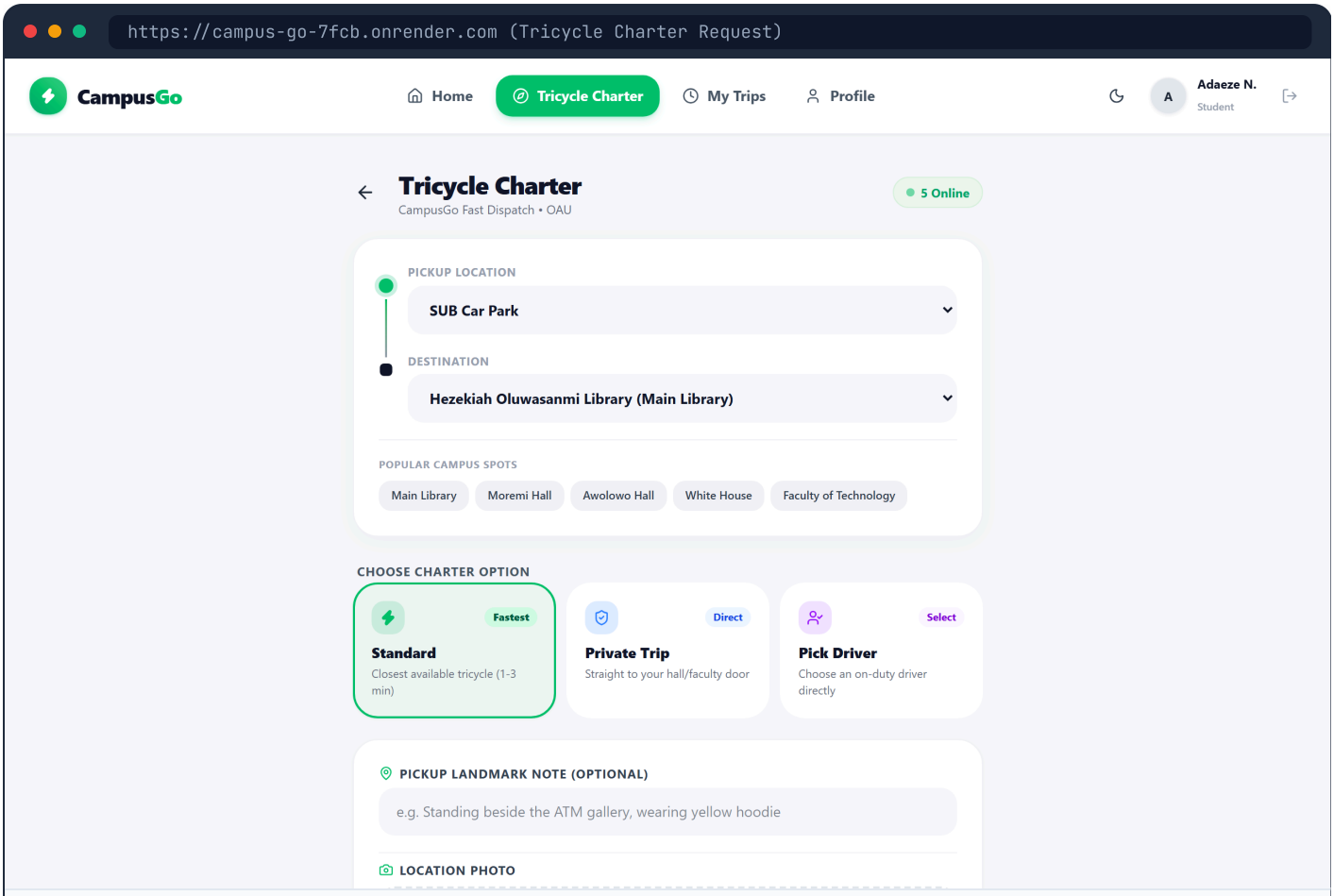
SMART ROUTING

Includes over 18 pre-configured campus landmarks across academic and residential zones (Senate Building, Spider House, Moremi Hall, PG College, Health Sciences, E.T.F. Hall, Fajuyi, Awolowo Hall).

Transparent Fare Estimation & Live Status (FR-26, FR-28)

FARE PREDICTABILITY

Calculates estimated fare quotes based on standard distance bands (₦300 - ₦600) and displays live lifecycle statuses: `Pending` → `Accepted` → `In Progress` → `Completed`. Students see driver identity, phone call shortcut, and vehicle registration upon acceptance.



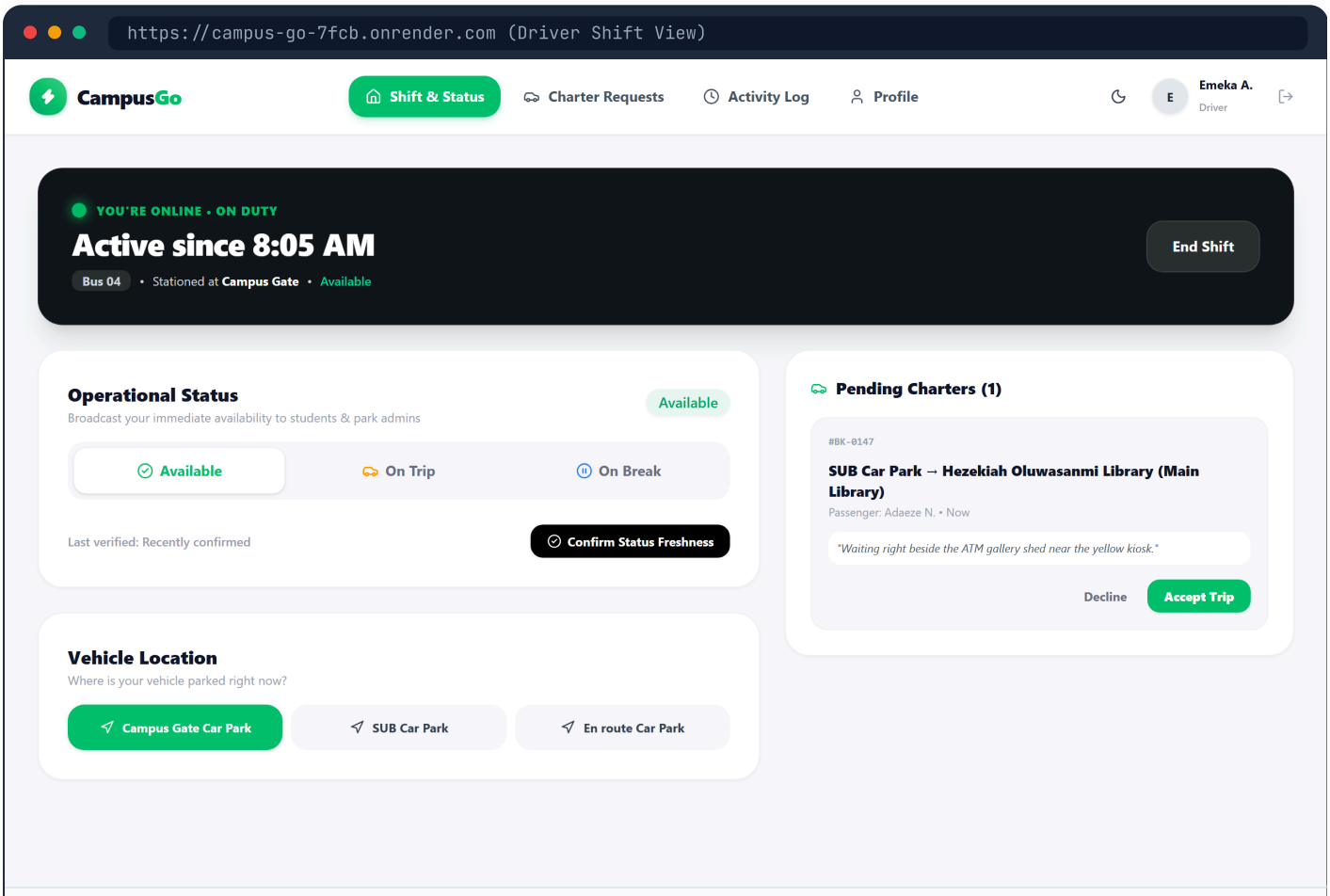
Screenshot 2: Tricycle charter booking screen with campus landmark selector, pickup time options, and fare calculator.

3. Driver Portal Core Functionalities

The driver interface provides licensed bus and tricycle operators with a distraction-free operational dashboard to control their shift availability, receive charter dispatches, and log trip progress.

3.1 Shift & Operational State Management (FR-05 to FR-09)

Status Field	Permitted Values	Operational Implication
Shift Status (FR-08)	NOT CHECKED IN | ON DUTY | SHIFT ENDED	Tracks daily work shift lifecycle. Driver cannot receive orders until checked in.
Operational Status (FR-07)	AVAILABLE | ON TRIP | ON BREAK | UNCONFIRMED	Indicates immediate passenger readiness. Taking a break pauses dispatch without ending shift.
Stale Detection (FR-09)	Auto-triggered after 2 hours without update	Flags driver as unconfirmed to prevent misleading students, without forcefully ending shift.



Screenshot 3: Driver Shift Management dashboard with one-touch shift check-in, operational status toggles, and live incoming requests stream.

3.2 Real-Time Charter Dispatch Stream (FR-27 to FR-29)

Incoming Request Cards: Displays student pickup point, target destination, passenger name, and estimated fare earnings.
One-Tap Accept / Decline: Accepting a request locks the booking immediately, preventing double-assignment (FR-28).
Active Trip Lifecycle: Prompts driver through sequential steps: `Start Trip` → `Complete Trip`.
Daily Earnings & Statistics: Tracks completed trips, shift hours, and revenue collected today (FR-30).

4. Park Administrator Portal Core Functionalities

Park administrators oversee operational execution at their assigned terminal (Campus Gate Car Park or SUB Car Park), dispatch fleet departures, and act as operational proxies for drivers without smart mobile devices.

4.1 Terminal Operations & Non-Smart Driver Proxy (FR-05, FR-10, FR-36)

"On-Behalf" Driver Management (FR-05)

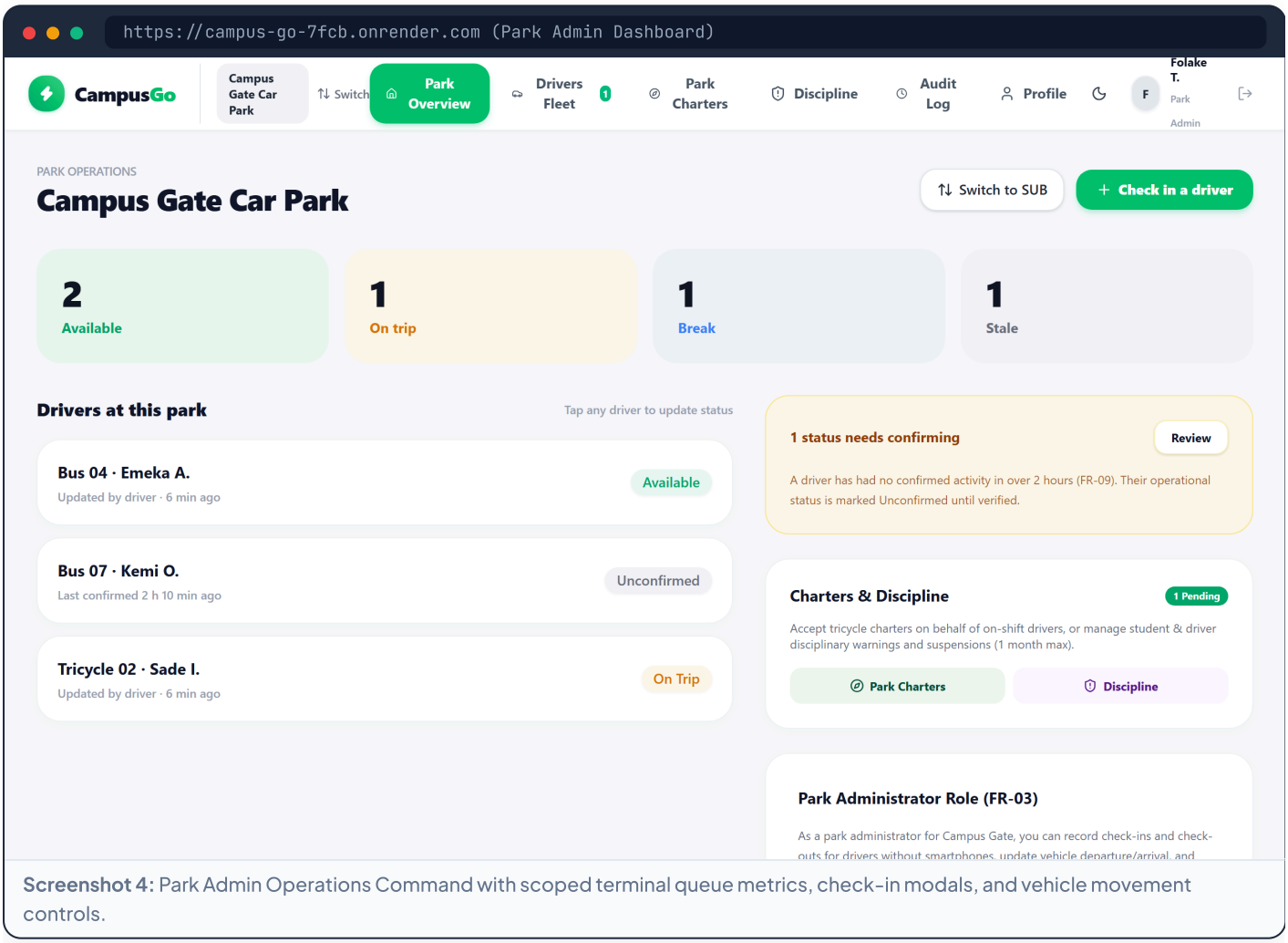
DIGITAL INCLUSIVITY

Drivers without smartphones or internet access report physically to the park admin booth. The Park Admin checks them in, logs departures, and updates operational statuses on their behalf. The audit log explicitly attributes these updates as `Updated by Admin on behalf of Driver` (FR-10).

Cross-Park Departure & En-Route Marking (FR-15, FR-17)

TRANSIT INTEGRITY

When a bus departs Campus Gate for SUB, the park admin marks the vehicle as `En Route`. This removes it from Campus Gate's available inventory without prematurely counting it as arrived at SUB until verified.



Screenshot 4: Park Admin Operations Command with scoped terminal queue metrics, check-in modals, and vehicle movement controls.

4.2 Disciplinary Actions (Park Admin Tier – FR-31)

Formal Disciplinary Warnings: Park Admins can issue formal written warnings with mandatory recorded reasons (e.g. queue jumping, fare disputes).

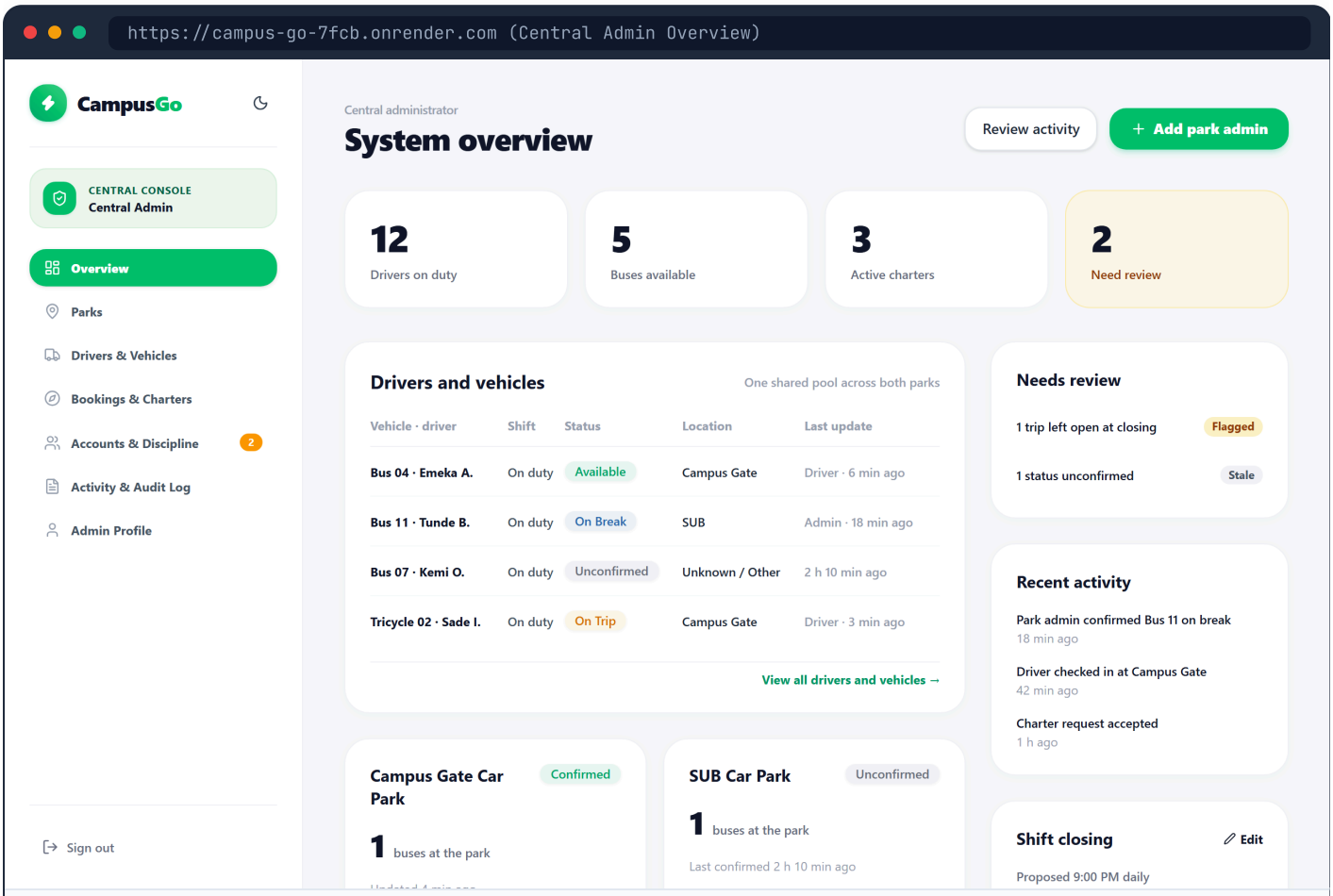
Capped 30-Day Suspension Authority: Park Admins may issue temporary operational suspensions strictly capped at 30 days. Indefinite or permanent suspensions require Central Admin escalation.

5. Central Administrator Portal Core Functionalities

The Central Administrator possesses university-wide authority over all parks, fleet assets, user accounts, and disciplinary policies, backed by an immutable activity audit trail.

5.1 Executive Overview & Master Registry (FR-32, FR-33)

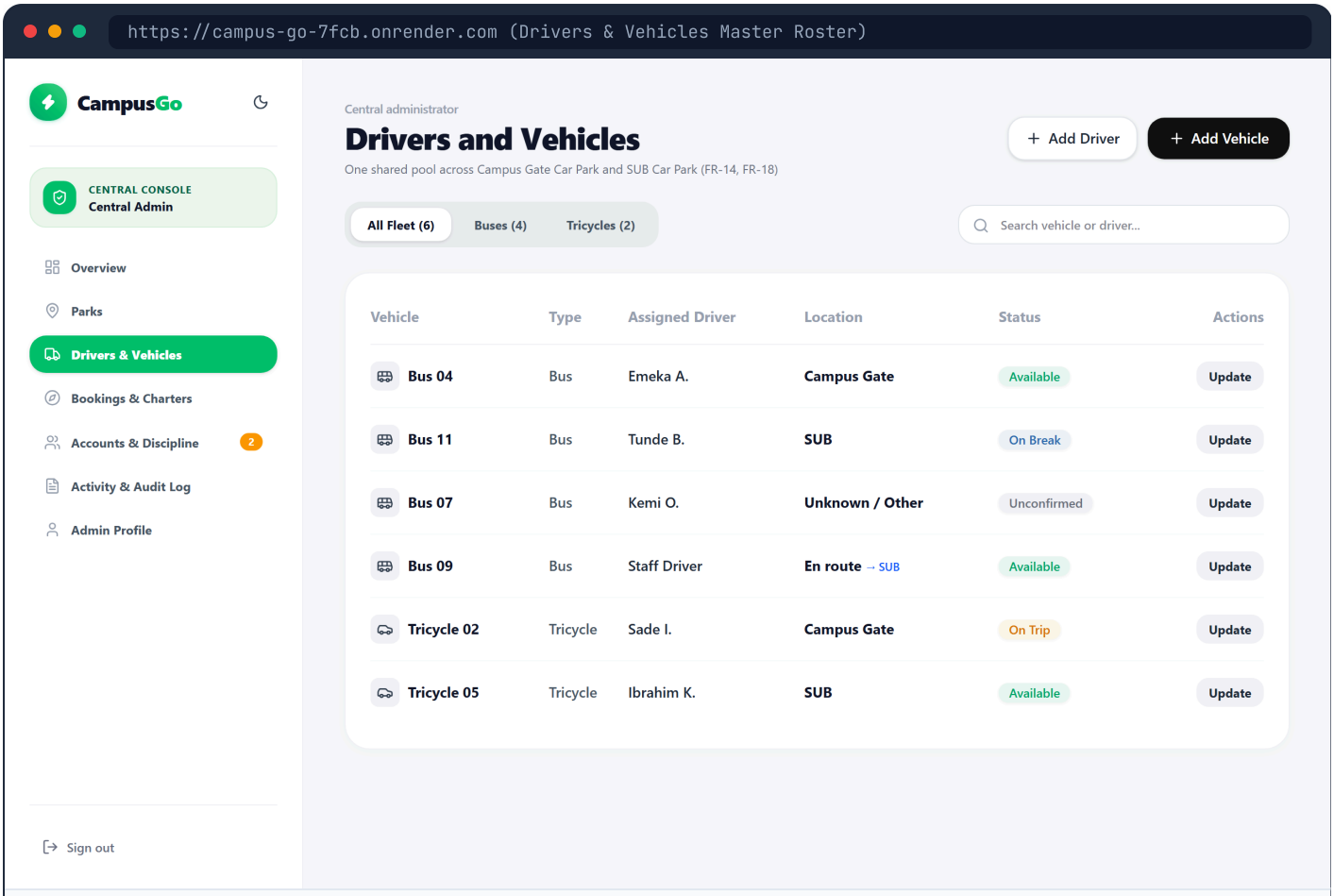
Provides high-level KPIs across the entire OAU campus transport network: active on-duty drivers, total fleet operational distribution, daily charter bookings, and unresolved trip alerts.



Screenshot 5: Central Admin Executive Dashboard with real-time fleet analytics, terminal balances, and quick administrative actions.

5.2 Fleet & Driver Master Management (FR-14, FR-32)

Central administration manages driver onboarding, verification, vehicle allocation (Buses and Tricycles), and status overrides across both Campus Gate and SUB Car Parks.



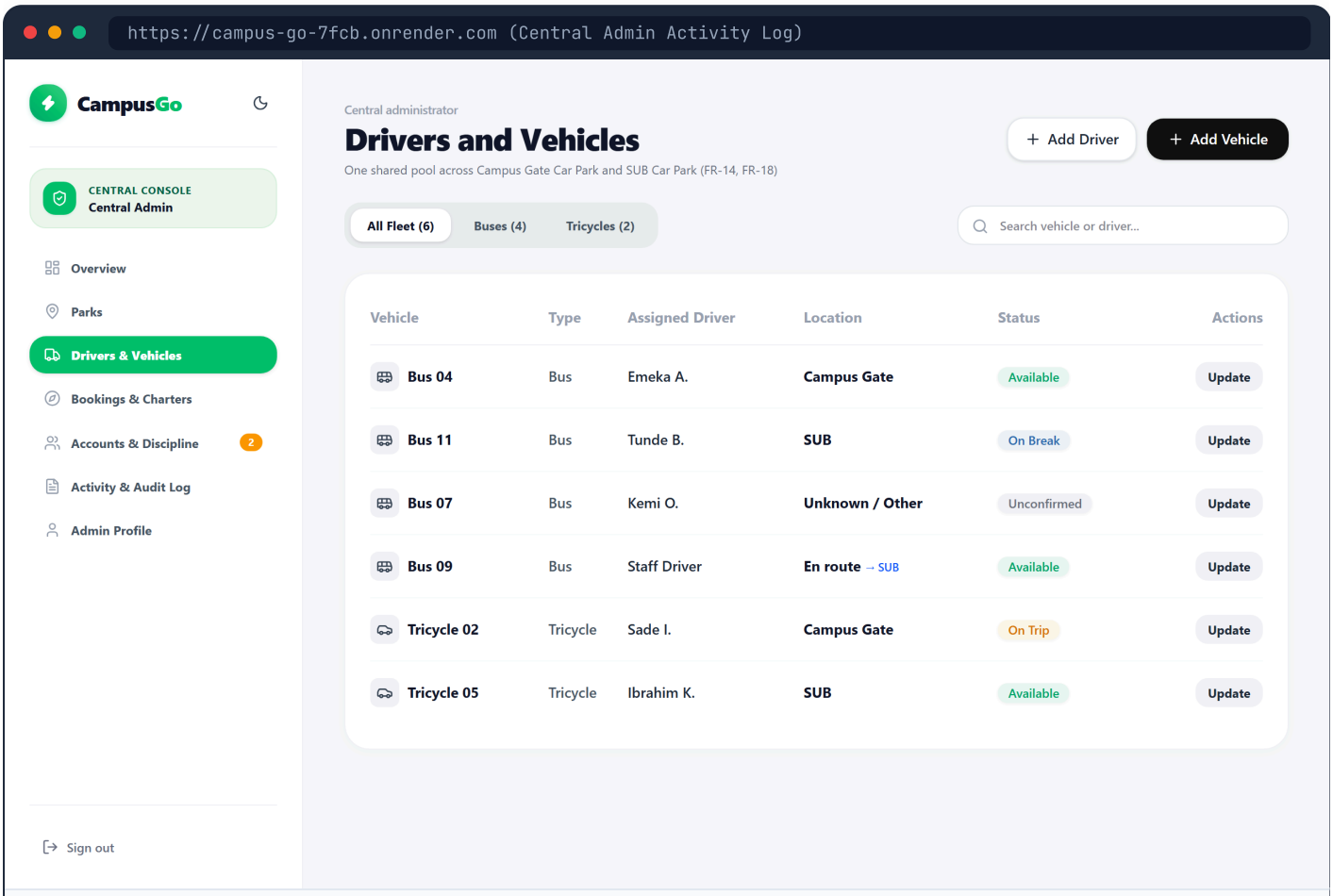
Screenshot 6: Complete driver and vehicle inventory management, credential validation, and vehicle reassignments.

6. Activity Auditing & Automated Disciplinary Engine

To guarantee accountability, every state transition in CampusGo is logged into an immutable audit trail, while disciplinary penalties are governed by an automated lifecycle engine.

6.1 Real-Time Activity Log (FR-10, FR-35, FR-36)

Every driver check-in, status adjustment, departure dispatch, disciplinary penalty, and record correction is captured with UTC timestamp and actor identity.



Screenshot 7: Real-time activity log showing granular audit entries for status updates, disciplinary penalties, and record corrections.

6.2 Disciplinary Authority Matrix & Automated Expiry (FR-31)

Disciplinary Action	Park Admin Authority	Central Admin Authority	Automated Resolution
Issue Warning	Yes (Mandatory reason required)	Yes (Mandatory reason required)	Permanent record on driver profile
Temporary Suspension	Yes (Strictly capped at 1 to 30 days)	Yes (Any duration)	**Auto-Expiry Engine:** System automatically reinstates driver when suspension date elapses.
Indefinite Suspension	No (Authority restricted)	Yes (Full authority)	Requires manual reactivation by Central Admin
Override / Revocation	Cannot override Central decisions	Can override or lift any Park Admin penalty	Immediate restoration of driver access

7. User Accounts & Engineering Innovations

7.1 Pre-Configured Demonstration Accounts

All roles can be evaluated on the live production site using the verified credentials below (Password: `password123`):

Role	Account Name	Email Address	Assigned Scope / Vehicle	Password
CENTRAL ADMIN	Central Admin	`centraladmin@university.edu`	All Parks Oversight	`password123`
PARK ADMIN	Folake T.	`folake@university.edu`	Campus Gate Car Park	`password123`
PARK ADMIN	Samuel D.	`samuel@university.edu`	SUB Car Park	`password123`
BUS DRIVER	Emeka A.	`emeka@university.edu`	Bus 04 (On Duty)	`password123`
TRICYCLE DRIVER	Sade I.	`sade@university.edu`	Tricycle 02 (On Duty)	`password123`
STUDENT	Adaeze N.	`adaeze@university.edu`	Matric: CSC/2021/042	`password123`
SUSPENDED STUDENT	Ibrahim M.	`ibrahim.m@university.edu`	Matric: ECN/2021/019 (Suspended 14d)	`password123`

7.2 Technical Optimizations & Architecture

Client-Side Image Compression (>97% Reduction)

PERFORMANCE

Implemented custom HTML5 Canvas image pre-processing (`src/lib/imageCompression.ts`) that automatically scales avatars to max 800px and 75% quality before upload. Reduces typical 5MB mobile photos to under 70KB, dramatically reducing mobile data usage on campus cellular networks.

Non-Blocking Toast Notification System

MODERN UX

Replaced intrusive native browser `alert()` popups with custom, animated, non-blocking toast notifications positioned at the bottom of the screen with smooth fade-out animations.

Zero-Auto-Login Session Security

SECURITY

New visitors land directly on the clean Sign In / Sign Up portal without defaulting into any user profile. Sessions persist in `localStorage` only while active, and logging out completely clears session tokens, requiring full re-authentication to enter the application.